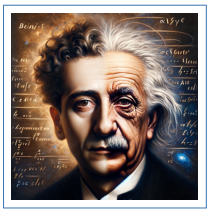
Xianquan Yan

Curriculum Vitae

Department of {Physics, Computer Science}
National University of Singapore
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Education

2023–present **PhD Candidate, {Physics, Computer Science}**, *National University of Singapore*, Singapore.
Supervisors: Lee Ching Hua and Kenji Kawaguchi
Research Interests: Topological Non-Hermitian Physics, AI + Science, Complex Systems, Quantum Information Science
GPA: 5.00/5.00

2019–2023 **BS, Physics**, *Sichuan University*, Chengdu, China.
GPA: 3.96/4.00 (Ranking 1/285)

2022 **Visiting Student (BPIE), Physics**, *University of California, Berkeley*, CA, USA, *GPA: 4.00/4.00*.

2022 **Summer Institute, Physics**, *University of Oxford*, Oxford, UK, *GPA: 90/100*.

Publications

*Equal contribution. †Corresponding author(s).

First / Co-first & Corresponding Author

01 **Topological classification through knotted graphs: Fermi surface dispersions and Lifshitz transitions**
Hakan Akgün*†, **Xianquan Yan***†, and Ching Hua Lee†
[Equal contribution; co-first authors listed alphabetically.]
arXiv preprint arXiv:2609.13390 · 2026 · *Under review at* **Nature Physics**

02 **Observing the magic Mpemba effect in localized dynamics on a digital quantum computer**
Han-Ze Li*, **Xianquan Yan***, Yi-Rui Zhang, Jian-Xin Zhong, Shuo Liu, and Ching Hua Lee
arXiv preprint arXiv:2610.08769 · 2026 · *Under review at* **Physical Review Letters (PRL)**

03 **HSG-12M: A Large-Scale Benchmark of Spatial Multigraphs from the Energy Spectra of Non-Hermitian Crystals**
Xianquan Yan†, Hakan Akgün, Kenji Kawaguchi, N. Duane Loh†, and Ching Hua Lee†
International Conference on Learning Representations (ICLR) 2026
CCF A (2026) ICORE A* (2026) Impact Factor* 56.04 (2026)
Media Coverage: [NUS Science] [Sichuan University Physics]

* Estimated conference impact score from CS Conference Impact Factors

04 **Non-Hermitian Activation of Quantum Mpemba Effect in Dipole-conserving Systems**
Yi Qin†, Han-Ze Li, Chun-Hui Liu, **Xianquan Yan**†, and Ching Hua Lee†
2026 · *Under review at* **Nature Communications**

05 **Close and Sort: Exact Maximum-Likelihood Aggregation of Cyclic Rank-Two Preferences**
Kehan Liu, Zhaoyun Chen, and **Xianquan Yan**†
ACM International Conference on Web Search and Data Mining (WSDM) 2027 · *Accepted*
CCF B (2026) ICORE A (2026) Impact Factor* 15.50 (2026)

06 **When Greedy Is Sorting: Exact and Certified Ideal Rankings for Novelty Metrics**
Kehan Liu, Zhaoyun Chen, and **Xianquan Yan**†
ACM International Conference on Web Search and Data Mining (WSDM) 2027 · *Accepted*
CCF B (2026) ICORE A (2026) Impact Factor* 15.50 (2026)

07 **Classifying the Graph Topology of Non-Hermitian Energy Spectra with Graph Transformer**
Xianquan Yan, Hakan Akgün, Kenji Kawaguchi, N Duane Loh, and Ching Hua Lee
AAAI 2026 XAI4Science Workshop

08 **Multi-Modal Classification of Extended 1D Cellular Automata**
Zhaoyun Chen*, **Xianquan Yan***, Hakan Akgün, N Duane Loh, and Ching Hua Lee
AI4X — Accelerate Conference 2026

09 **Automated Metric Discovery: Navigating Quantum Geometry with Symbolic Regression**
Xianquan Yan†
AI4X 2025 International Conference

10 **Precise Identification of Topological Phase Transitions with Eigensystem-Based Clustering**
Xianquan Yan and Jian-Song Pan
AI4X 2025 International Conference

Co-author

11 **Deterministic scale-invariant dynamics in a logistic Game-of-Life model**
Hakan Akgün, **Xianquan Yan**, Tamer Taşkıran, Muhamet Ibrahimi, Ching Hua Lee, and Seymur Jahangirov
Communications Physics · 2026
XinRui Tier 1 / Top (2026) JCR Q1 (2025) Impact Factor 5.5 (2025)

12 **KnottedGraph: Scalable knotted-graph topology for scientific and mathematical discovery**
Hakan Akgün, **Xianquan Yan**, Kehan Liu, Zhaoyun Chen, and Ching Hua Lee
arXiv preprint arXiv:2609.31152 · 2026

13 **Practical Error Suppression and Mitigation for Reliable Quantum Computing**
Han-Ze Li, Mengjie Yang, **Xianquan Yan**, Dax Enshan Koh, Ching Hua Lee, and Ruizhe Shen
arXiv preprint arXiv:2608.20453 · 2026 · *Under review at* **Advanced Quantum Technologies**

Experience

2024–2026 **Teaching Assistant**, *National University of Singapore*, Singapore.
- (PC5253) Complex Systems Analysis and Modelling
- (AIS5102) Practical Machine Learning for Scientific Discovery
- (AIS5204) AI in Condensed Matter Physics

2022 **Student Research Assistant**, *Lawrence Berkeley National Laboratory*, Berkeley, CA, USA.
Supervisor: Benjamin Nachman

Academic Service

2026 Reviewer, *Communications Physics*, Nature Publishing Group UK London
2026 Reviewer, *NeurIPS 2026*
2026 Reviewer, *ICLR 2027*

Codebases and Datasets

`KnottedGraph`: Python framework for constructing, visualizing, and analyzing knotted graphs, including scalable Yamada polynomial evaluation.
[https://github.com/HakanAkgn/KnottedGraph]

`poly2graph`: Python package for automatic Hamiltonian spectral graph construction from characteristic polynomials and Bloch Hamiltonians.
[https://github.com/sarinstein-yan/poly2graph]

`HSG-12M`: Codebase for downloading, processing, subsetting, and benchmarking the HSG-12M spatial multigraph dataset with NetworkX and PyTorch Geometric.
[https://github.com/sarinstein-yan/HSG-12M]

HSG-12M (DATA): Harvard Dataverse dataset of spatial multigraphs extracted from the energy spectra of non-Hermitian crystals.
[https://doi.org/10.7910/DVN/PYDSSQ]

`ClusterAnalyzer`: Python library for analyzing critical behavior, self-similarity, and fractal dimensions of clusters in 2D cellular automata.
[https://github.com/HakanAkgn/ClusterAnalyzer]

Honors and Awards

2021 **China National Scholarship** (Top 0.2%), The Ministry of Education of the P.R. China

2020 **China National Scholarship** (Top 0.2%), The Ministry of Education of the P.R. China

2022 **1st Prize, Scholarship of *Bajian* program for fundamental disciplines 2.0**, The Ministry of Education of the P.R. China

2022 **1st Prize, CASC Scholarship**, China Aerospace Science and Technology Corp

2012 **Gold, Asia International Mathematical Olympiad Open Contest**, A.I.M.O. Union

2019 **Provincial Excellent Student Leader**, The Education Department of Sichuan Province

2020 **1st Prize** and **Best Demeanor Award**, South-West Undergraduate Physics Tournament, Sichuan Physical Society

2020 **2nd Prize, National English Competition for College Students**, Advisory Board for College Foreign Language Teaching

2026 **Faculty of Science PhD Conference Award (FPCA)**, Faculty of Science, National University of Singapore

Extracurricular Activities

2013–2023 **Football (Soccer)**, *Captain, Left-Winger*, School Team.

2019–2021 **Debater**, *Debate Team (English) of Sichuan University*.

2019–2021 **Debater**, *Debate Team of College of Physics, Sichuan University*.

2021–2023 **Director**, *Sichuan University Mathematica Society*.

2019–2021 **Director**, *Sichuan University English Association*.

Calligraphy and Marathon Running.